

Materials

- Zip-lock bags
- Spatula, stirring rod
- Measuring cylinder
- Beakers
- 4% Borax Solution*
- PVA glue
- Food colouring
- Safety Glasses
- Plastic gloves

* (4% borax: add 40 g of sodium tetraborate ($\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$) to 1 litre of warm water. Stir well to mix. Some solid may remain undissolved.)

Optional : black plastic, plastic aprons (to protect benches and clothes)

Safety

Do not let students eat the slimes that are produced.

Wear gloves when mixing and handling the slime or prepare in zip-lock bags.

Safety glasses required.

Do not dispose of any substances containing Borax down the sink. Please wrap and dispose in landfill.

Refer to your own school/state protocols for dealing with hazardous substances.

What to do

- 1 Pose the focus question: *What type of stuff is slime?* and elicit students' ideas.
- 2 Provide students with instructions on making the slime – a flow chart may assist with scaffolding the instructions. Remind students of the safety precautions.
- 3 Guide students through the suggested questions as they are making observations:
 - o *how would you describe what you made?*
 - o *what sort of substance is your slime?*
 - o *what are the differences between what you started with and what you ended up with?*
- 4 Students notebook and represent their understanding by recording observations and summarising: For example:
 - o *how would you describe the slime?*
 - o *what sort of substance is it?*
 - o *what did you think or feel about this activity?*
 - o *comparing everyone's Slimes – what do they have in common*
What is different?
- 5 Use this Activity to lead into discussions about grouping or classifying materials in (Activity 1.2).

Instructions for Making Slime

- 1 Mix 50ml of water and 50ml of PVA glue together to create a thick solution.
- 2 Add a drop or two of food colour and stir until the colour is evenly distributed.
- 3 Slowly add 20ml of the borax solution, while stirring. The 'slime' will quickly form around the stirring rod and can be removed from the beaker and kneaded to make it thicker and drier. Wear disposable gloves while handling the 'slime'.
- 4 An alternative method involves making the 'slime' in a zip-lock bag. Place the mixture of glue water and food colour into the bag. Seal the bag and mix the contents thoroughly. Then add the borax solution and mix thoroughly.

Hints for Success

- This is very messy!
- To overcome the messiness black plastic sheets could be used to cover the benches.
- A flow chart diagram showing the instructions is available on the *Science by Doing* Student CD-ROM.
- Glassware will be difficult to clean if you leave the PVA in it.
- **CAUTION:** Refer to your state and territory guidelines for using borax.
- Slime is revisited again in Activity 2.11 where students investigate changing the proportions of ingredients to produce slime with specific properties.
- Avoid detailed explanations at this stage— the idea is that the student will not know how to classify this substance. As they work through this unit they will develop their understanding of materials that are solid / liquid / gas or combinations of these.
- Refer to Professional Learning Resources - *Notebooking, Managing Inquiry Based Classrooms, Cooperative and Collaborative Work, Inquiry Based Learning*.

